

The Public Ledger: Fiscal Policy and Taxation

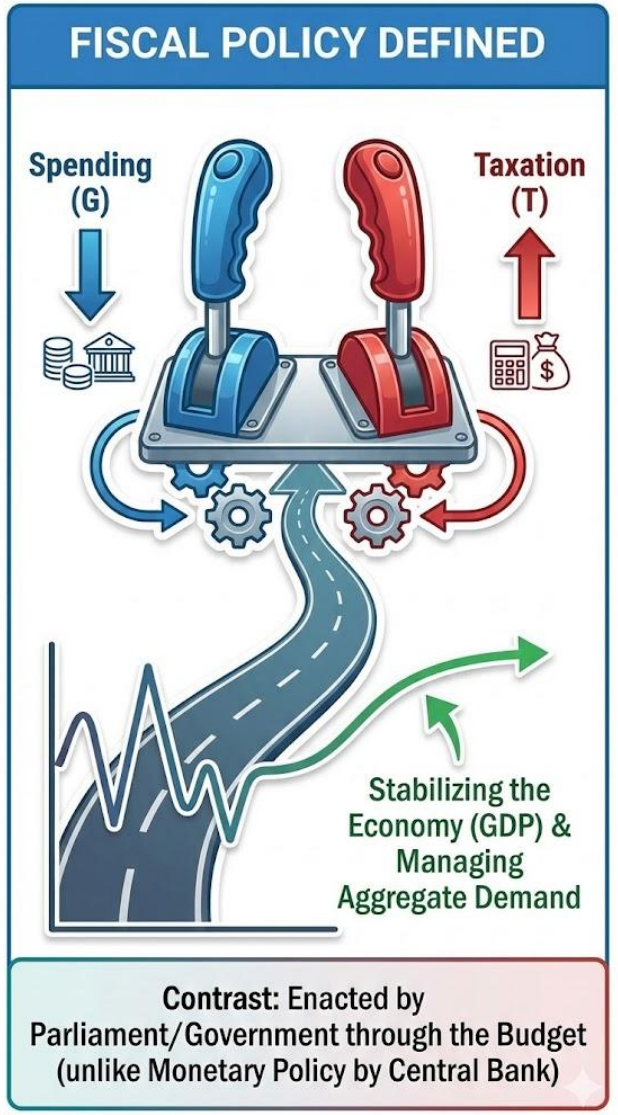
Fiscal Policy, Income Tax, Value Added Tax (VAT), Local Tax.

ITGE275 Economics

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Fiscal Policy Defined

- **Definition:** Fiscal Policy involves the use of government spending (***G***) and taxation (***T***) to influence the economy.
- **The Goal:** To manage aggregate demand, stabilize the business cycle (smoothing out booms and busts), and redistribute income.
- **Contrast:** Unlike Monetary Policy (Central Bank/Interest Rates), Fiscal Policy is enacted by the Parliament/Government through the Budget.

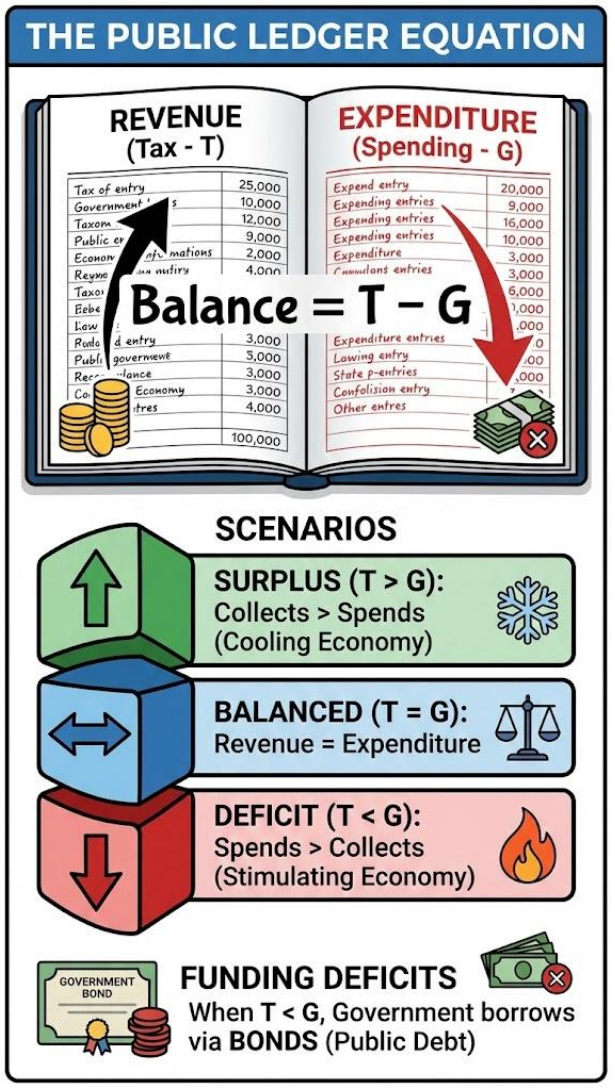


The Public Ledger Equation

- **The Budget Balance:**

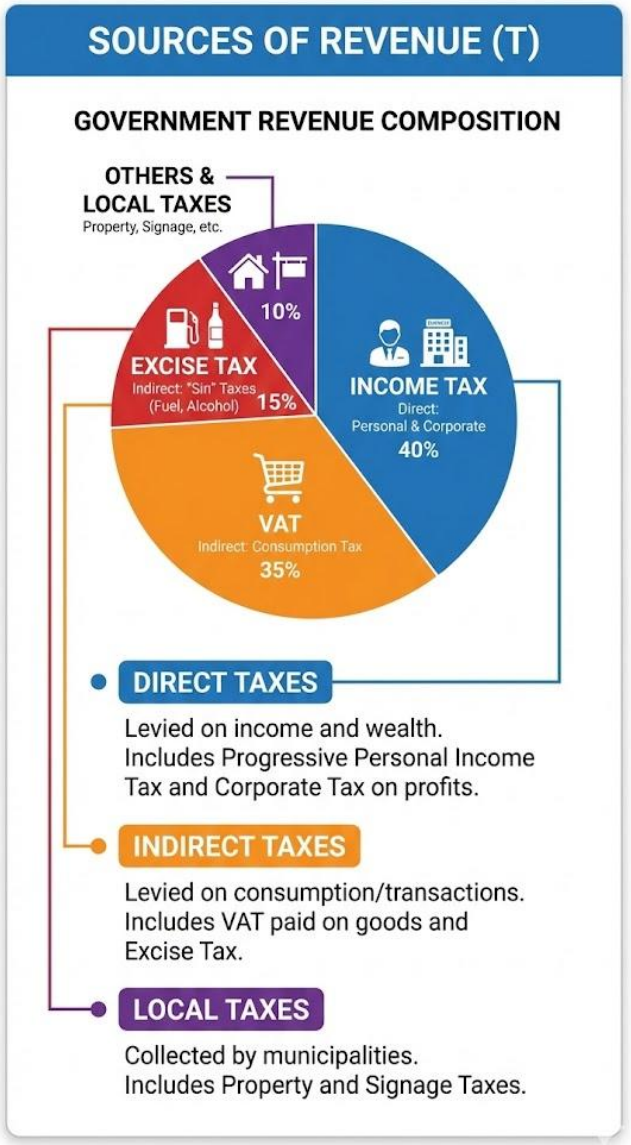
$$Balance = T - G$$

- **Scenarios:**
  - **Surplus (T > G):** The government collects more than it spends. (Cooling the economy).
  - **Deficit (T < G):** The government spends more than it collects. (Stimulating the economy).
  - **Balanced Budget (T = G):** Revenue equals expenditure.
- **Funding Deficits:** When T < G, the government must borrow money by issuing Bonds (Public Debt).



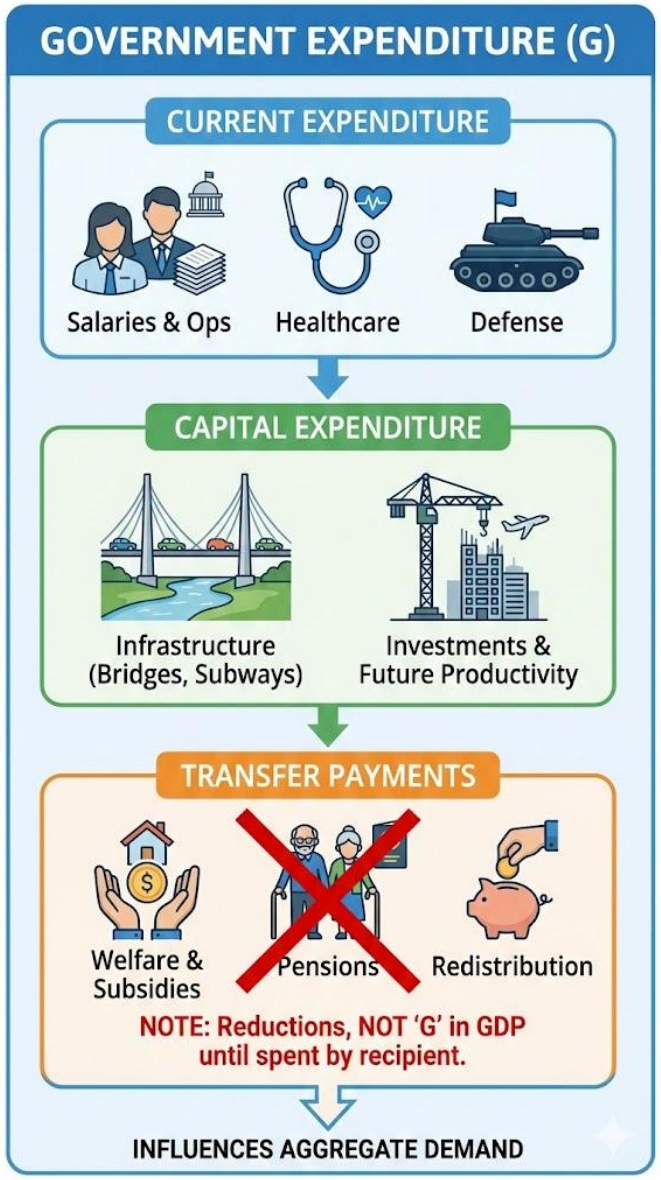
Sources of Revenue (T)

- **Direct Taxes:** Levied on income and wealth.
  - *Personal Income Tax:* Progressive tax on individuals.
  - *Corporate Tax:* Tax on company profits.
- **Indirect Taxes:** Levied on consumption and transactions.
  - *VAT (Value Added Tax):* Paid when buying goods.
  - *Excise Tax:* Sin taxes (Alcohol, Fuel).
- **Local Taxes:** Property and signage taxes collected by municipalities.



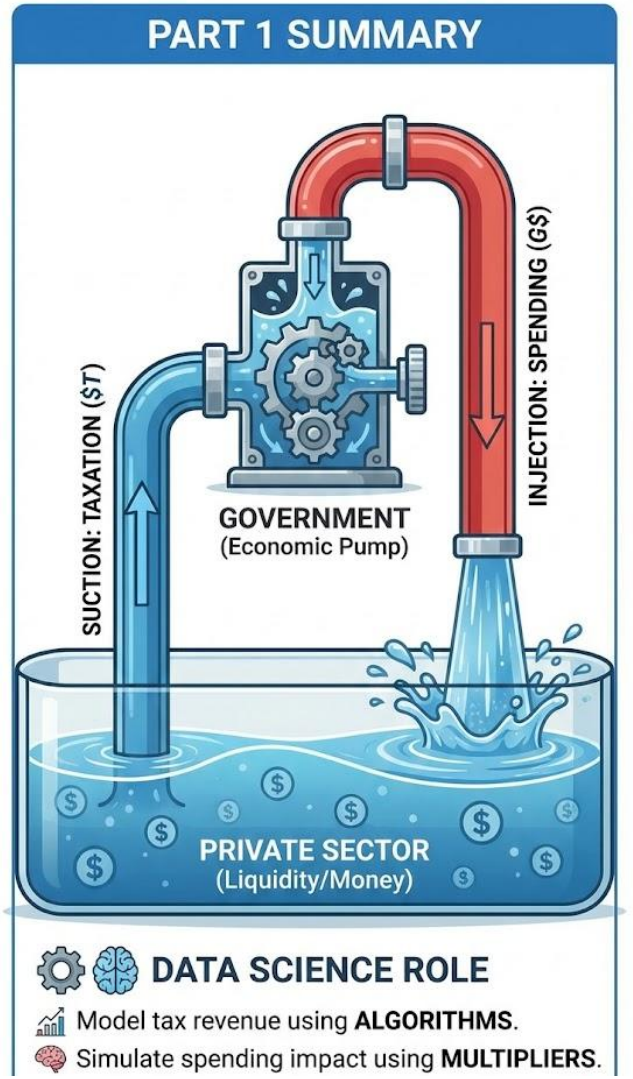
Government Expenditure (G)

- Where does the money go?
- **Current Expenditure:** Salaries of civil servants, healthcare operations, defense maintenance.
- **Capital Expenditure:** Investments in infrastructure (Subways, Bridges, Airports) that increase future productivity.
- **Transfer Payments:** Welfare, Pensions, Subsidies (Note: These redistribute money but don't count as "G" in GDP until the recipient spends it).



Part 1 Summary

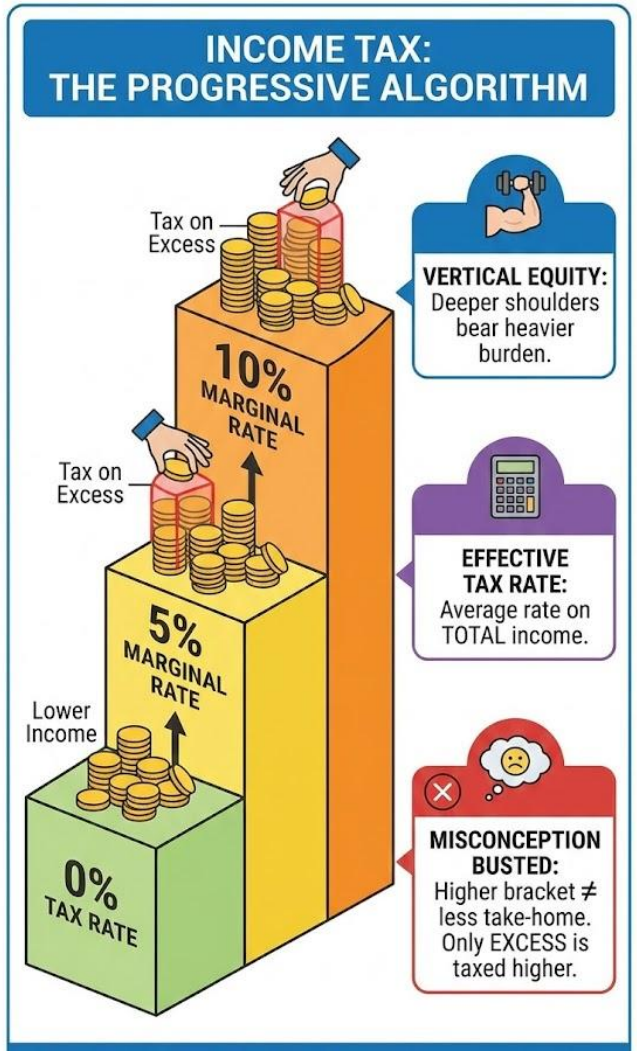
- The Government acts as a massive economic pump.
- It sucks in liquidity via Taxation (T).
- It injects liquidity via Spending (G).
- **Data Science Role:** We can model tax revenue using algorithms and simulate the impact of spending using multipliers.
- **Next:** How exactly is tax calculated?



Part 2: The Algorithms of Taxation

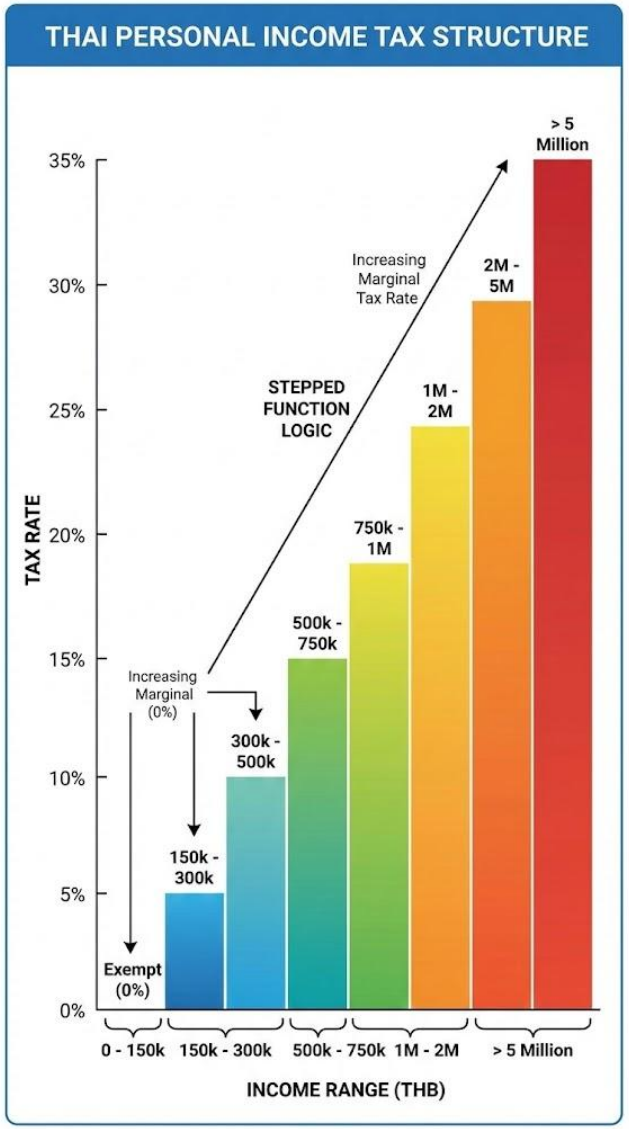
Income Tax: The Progressive Algorithm

- **Concept:** Vertical Equity. Those with deeper shoulders should bear a heavier burden.
- **Progressive Tax:** The tax rate increases as the taxable amount increases.
- **Marginal Tax Rate:** The tax percentage applied to the *last* dollar earned.
- **Effective Tax Rate:** The average rate you pay on your *total* income.
- **Misconception:** "Moving to a higher bracket means I take home less money." (False: You only pay the higher rate on the *excess* amount).



Thai Personal Income Tax Structure

- We use this structure in our Python code:
  - 0 - 150k: **Exempt (0%)**
  - 150k - 300k: **5%**
  - 300k - 500k: **10%**
  - ... up to **35%** for income > 5 Million.
- **Logic:** This creates a stepped function for tax liability.

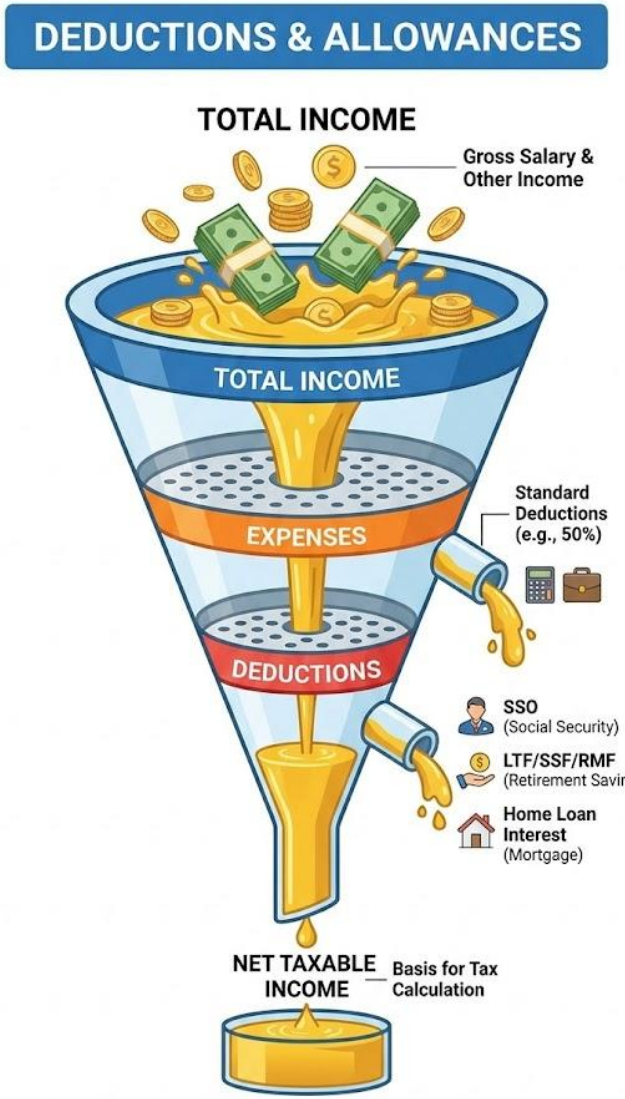


Deductions & Allowances

- Tax is calculated on **Net Taxable Income**, not Total Income.

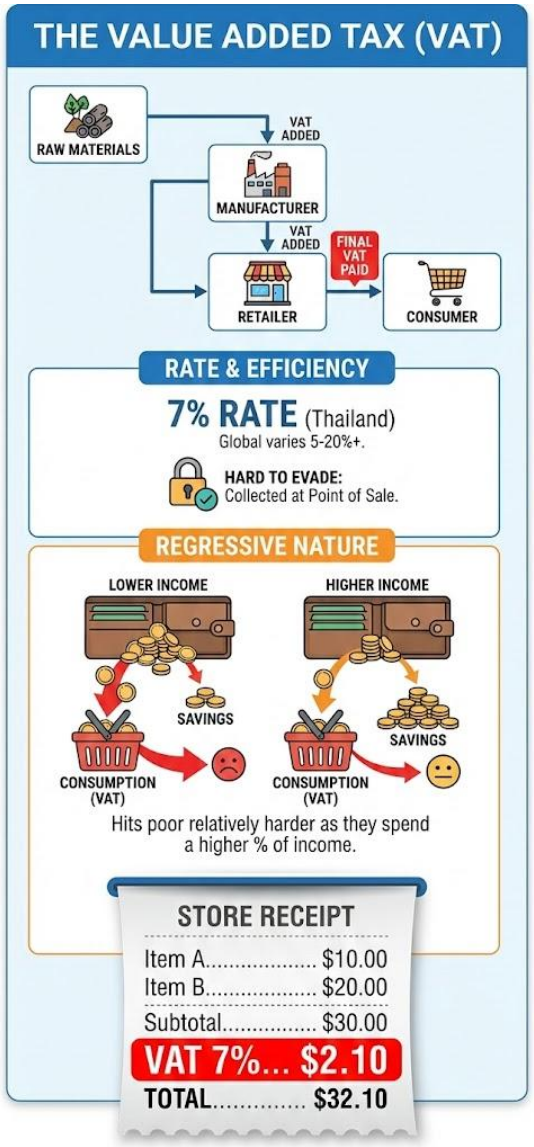
$$Net = TotalIncome - Expenses - Deductions$$

- **Expenses:** Standard deductions (e.g., 50% of salary).
- **Deductions:** Policy incentives to encourage specific behaviors:
  - *SSO:* Social Security.
  - *LTF/SSF/RMF:* Long-term savings for retirement.
  - *Interest:* Home loan interest.



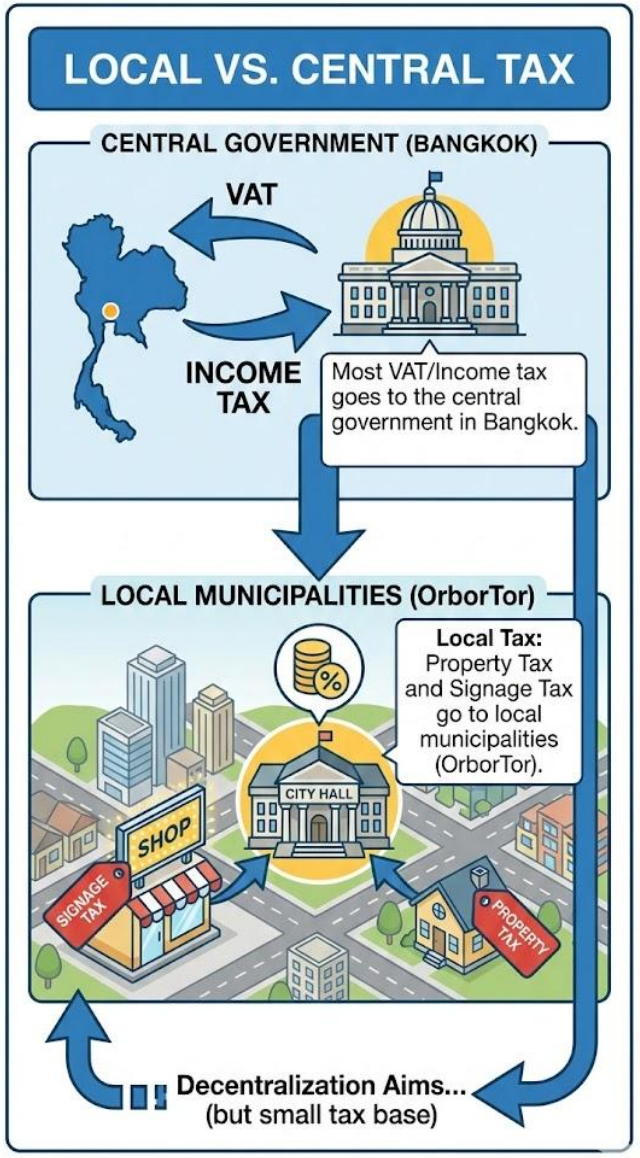
The Value Added Tax (VAT)

- **Concept:** A consumption tax placed on a product whenever value is added at each stage of the supply chain.
- **Rate:** Currently 7% in Thailand (Standard global rates vary from 5% to 20%+).
- **Efficiency:** Hard to evade because it is collected at the point of sale.
- **Regressive Nature:** Poor people spend a higher percentage of their income on consumption than rich people (who save more). Therefore, VAT hits the poor relatively harder.



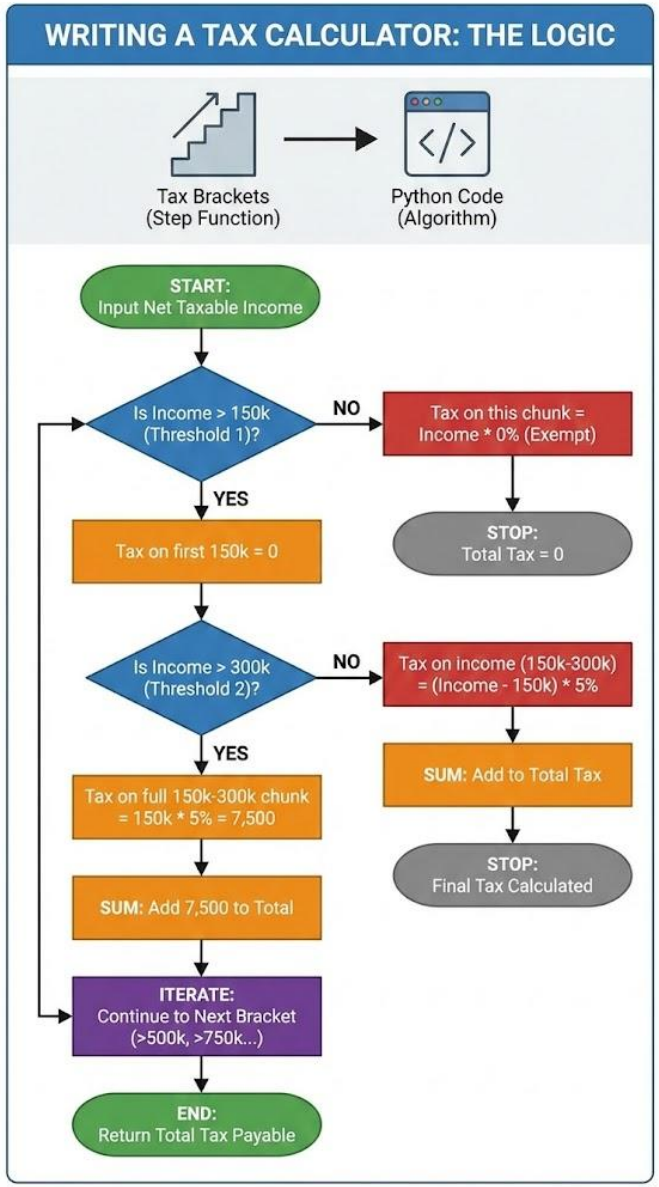
Local vs. Central Tax

- **Centralization:** Most VAT/Income tax goes to the central government in Bangkok.
- **Local Tax:** Property Tax and Signage Tax go to local municipalities (OrborTor).
- **Local Economics:** Decentralization aims to let local governments collect and spend revenue to fix local problems, but the tax base is often small compared to the central budget.



Computational Approach: Writing a Tax Calculator

- **The Challenge:** Converting the "Step Function" of tax brackets into Python code.
- **The Logic:**
  - Define brackets as a list of tuples: (Threshold, Rate).
  - Iterate through brackets.
  - Calculate income falling within each specific bracket.
  - Apply the specific rate to that chunk.
  - Sum the chunks.

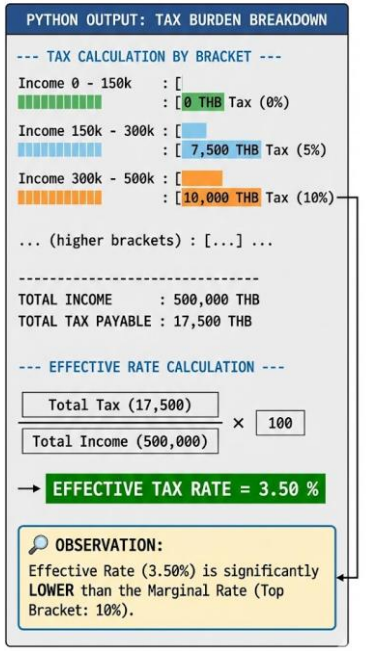


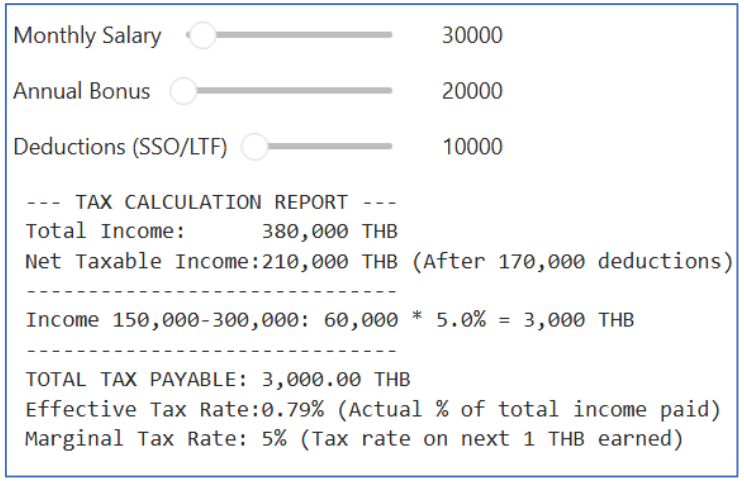
Visualizing the Tax Burden

- In the notebook, we output a breakdown.
- **Example:**
  - Income 0-150k: 0 Tax.
  - Income 150k-300k: 7,500 Tax.
- **Effective Rate Calculation:**

$$\frac{\text{Total Tax}}{\text{Total Income}} \times 100$$

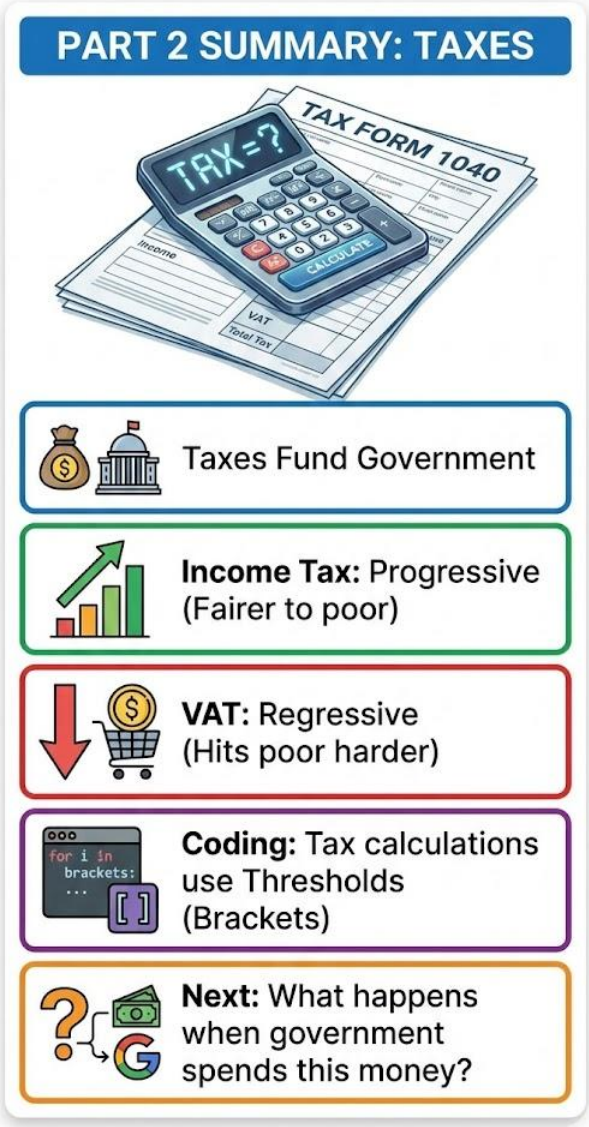
- **Observation:** The Effective Rate is always lower than the Marginal Rate (Top Bracket).





Part 2 Summary

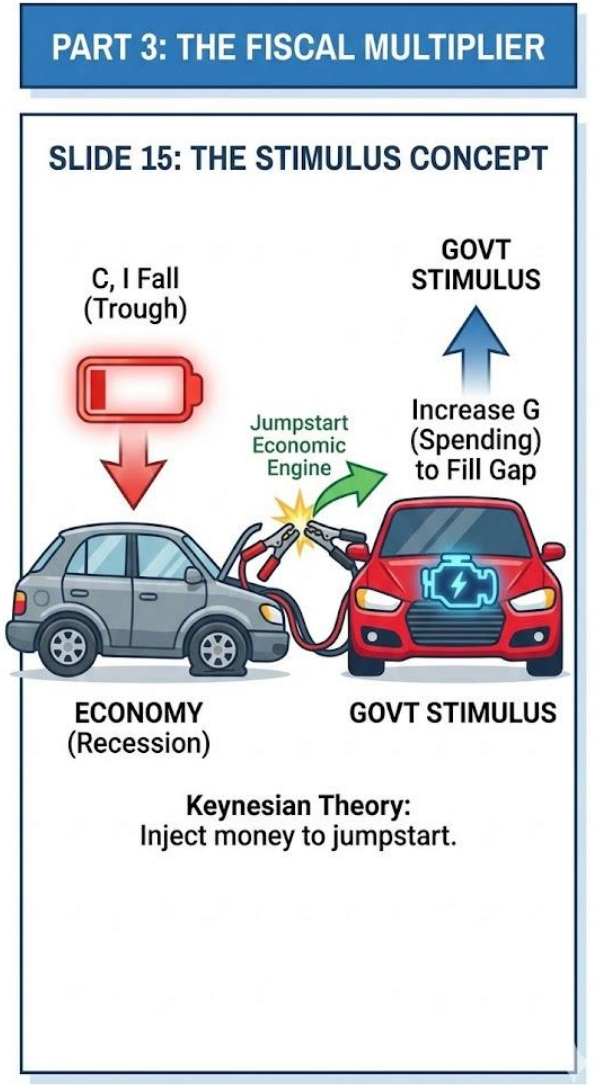
- Taxes fund the government.
- **Income Tax:** Progressive (Fairer to poor).
- **VAT:** Regressive (Efficient but hits poor harder).
- **Coding:** Tax calculations require iterating through thresholds (Brackets).
- **Next:** What happens when the government spends this money?



Part 3: The Fiscal Multiplier

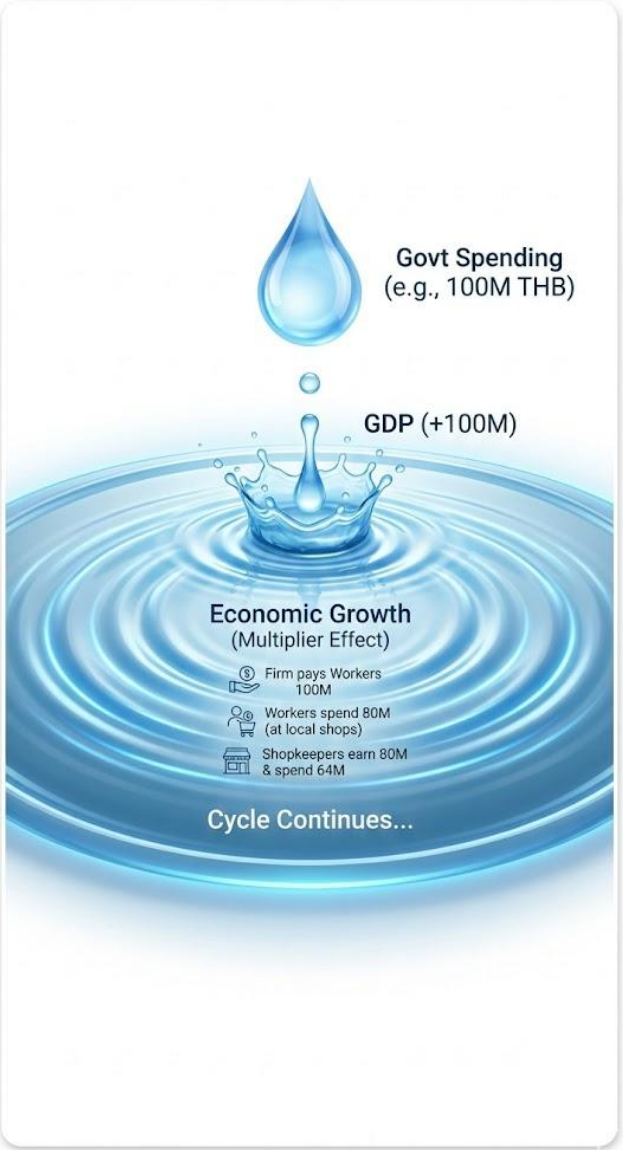
The Stimulus Concept

- When the economy enters a recession (Business Cycle Trough), Private Consumption (***C***) and Investment (***I***) fall.
- **Keynesian Theory:** The government should step in and increase Spending (***G***) to fill the gap.
- **Stimulus:** Injecting money to jumpstart the economic engine.



The Multiplier Effect Explained

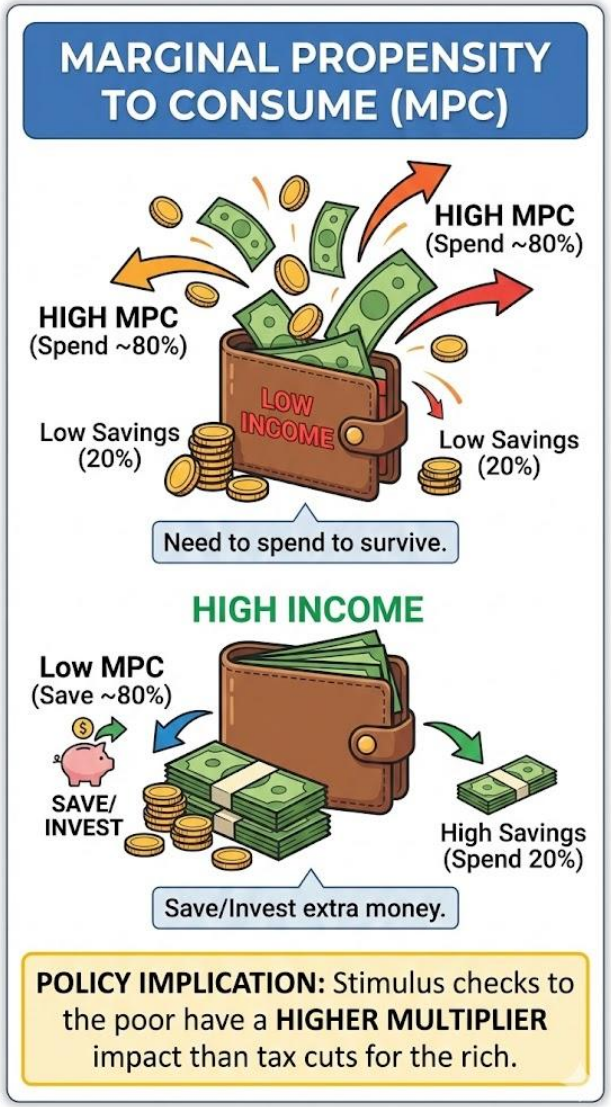
- **The Magic:** If the government spends 100 Million Baht, GDP grows by *more* than 100 Million.
- **The Mechanism:**
  - Govt pays Construction Firm 100M for a bridge.
  - Firm pays Workers 100M.
  - Workers spend 80M at local shops.
  - Shopkeepers earn 80M and spend 64M.
  - The cycle continues...



Marginal Propensity to Consume (MPC)

- **Definition:** For every extra $1 you earn, how much do you spend?
- **MPC = 0.8:** You spend 80 cents, save 20 cents.
- **Determinants:**
  - **Low Income:** High MPC (They need to spend almost everything to survive).
  - **High Income:** Low MPC (They save/invest the extra money).
- **Policy Implication:** Stimulus checks given to the poor have a higher multiplier impact than tax cuts for the rich.



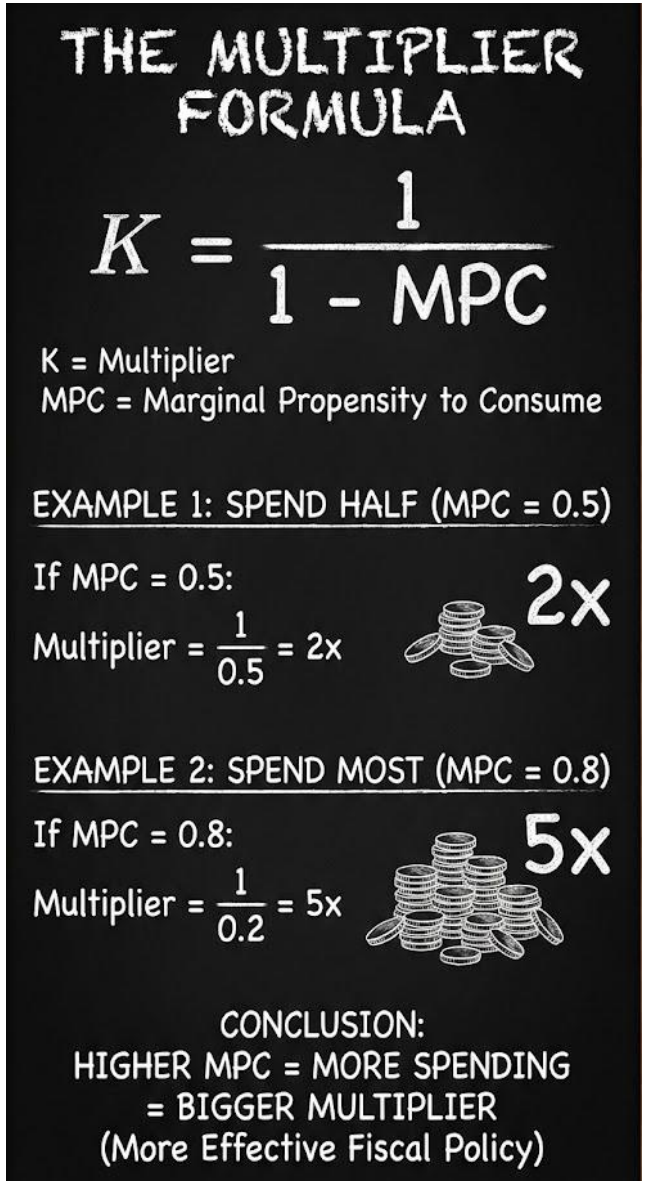


The Multiplier Formula

- **Theoretical Multiplier:**

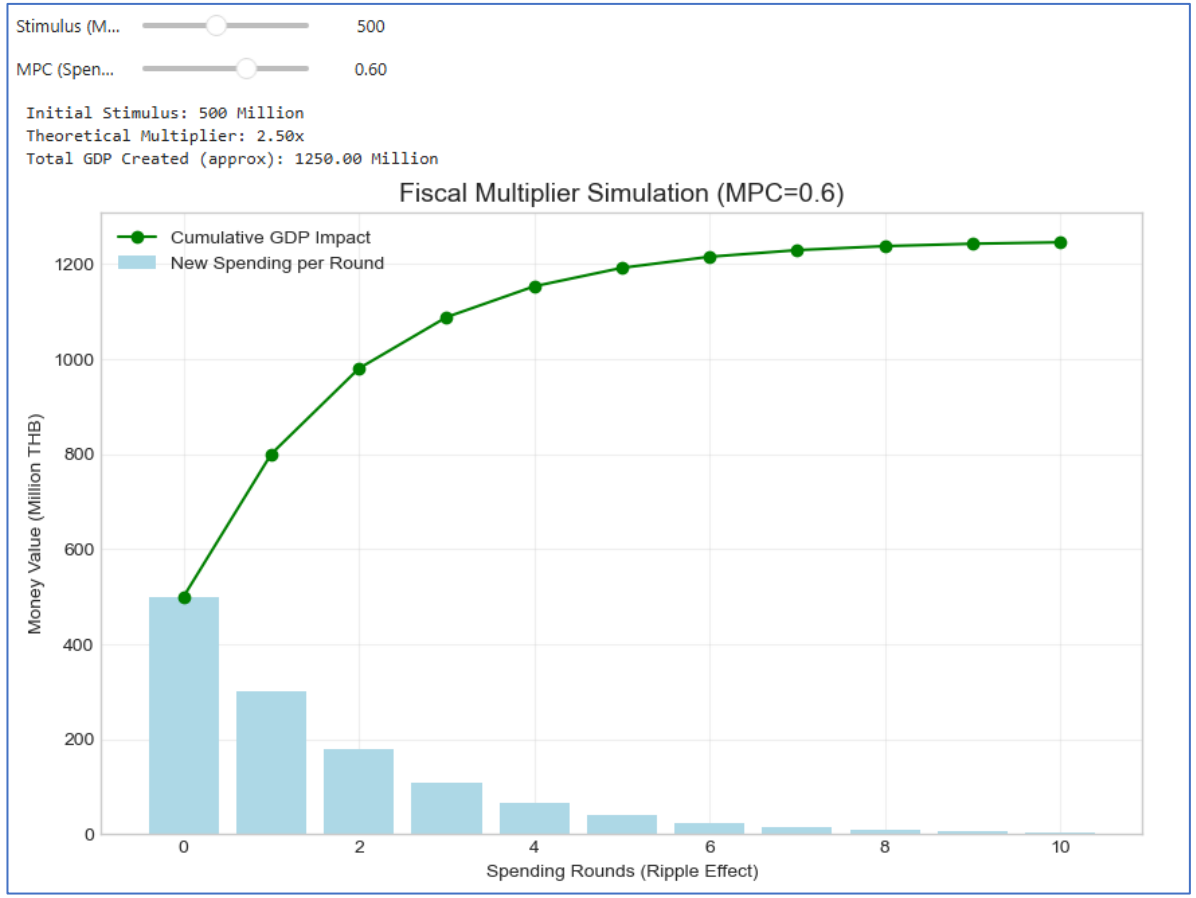
$$K = \frac{1}{1 - MPC}$$

- **Example:**
  - If MPC = 0.5 (Spend half): Multiplier = $1/0.5 = 2x$.
  - If MPC = 0.8 (Spend most): Multiplier = $1/0.2 = 5x$.
- **Conclusion:** The more people spend (rather than save), the more effective fiscal policy is.



Simulating the Multiplier in Python

- In the notebook, we build a fiscal_simulator.
- **Inputs:** Initial_Stimulus, MPC.
- **Process:** Loop through "Spending Rounds."
  - Round 1 = Stimulus
  - Round 2 = Round 1 * MPC
  - Round 3 = Round 2 * MPC
- **Output:** A cumulative chart showing how GDP grows round by round.



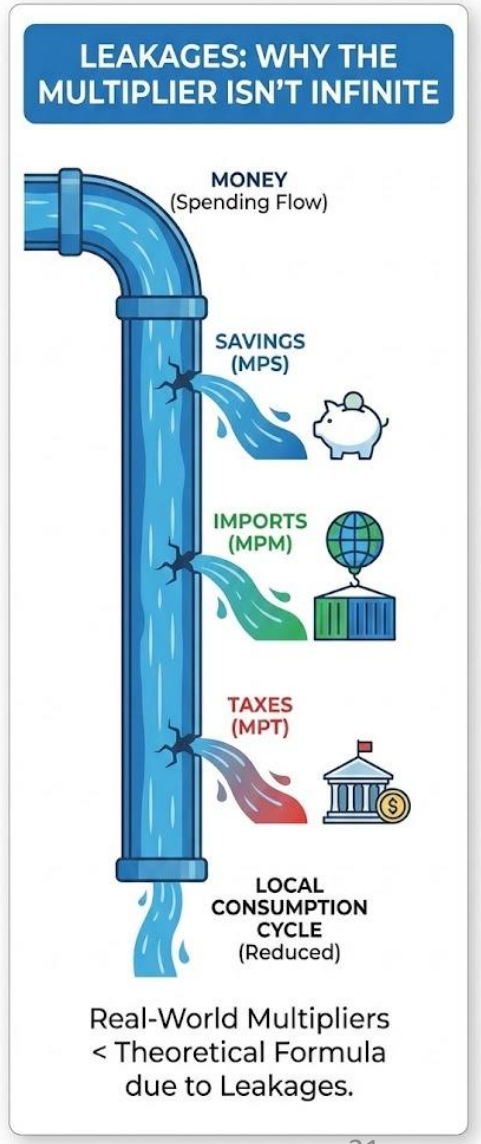
Case Study: The Digital Wallet

- **Scenario:** Government creates a digital token handout (e.g., 10,000 THB to citizens).
- **Restriction:** Must be spent within 6 months in the local district.
- **Goal:** To force a high MPC and keep the multiplier local.
- **Risk:** If people just substitute this money for what they would have spent anyway (saving their own cash), the MPC is effectively 0, and the multiplier is 1x.



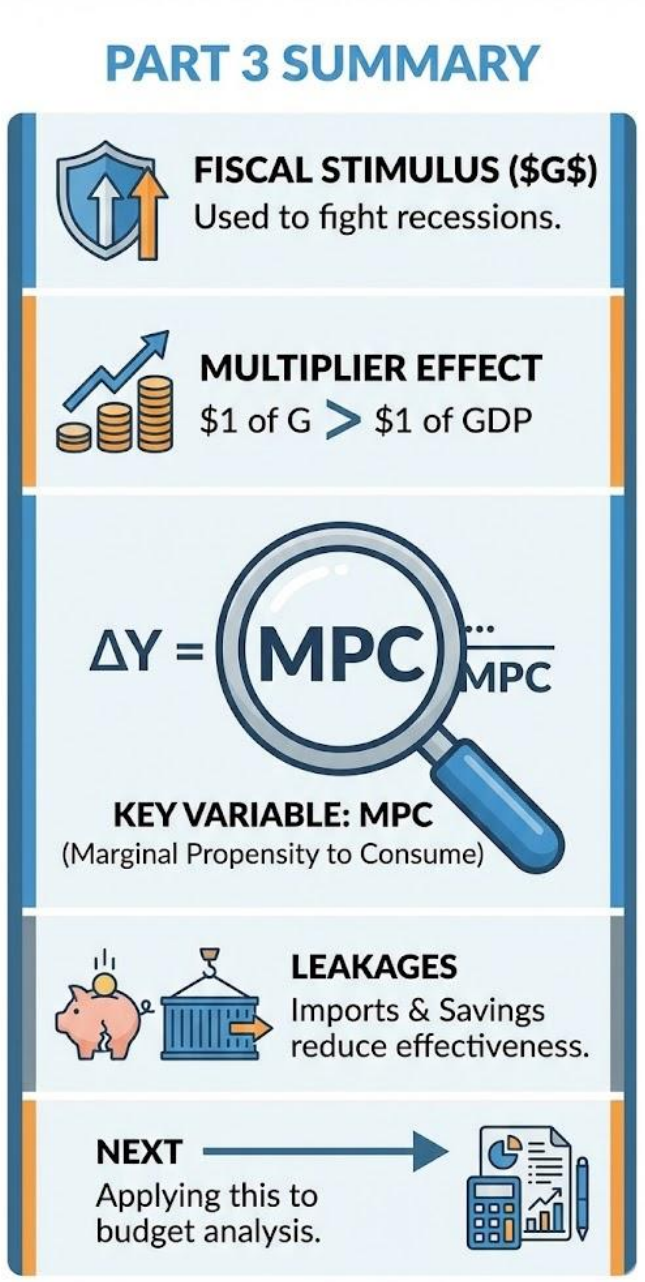
Leakages: Why the Multiplier isn't Infinite

- In the real world, multipliers are lower than the theoretical formula.
- **Leakages:** Money that leaves the local consumption cycle.
  - **Savings (MPS):** Money kept in the bank.
  - **Imports (MPM):** Buying an iPhone sends money to Apple (USA/China), not the local economy.
  - **Taxes (MPT):** Money goes back to the government.



Part 3 Summary

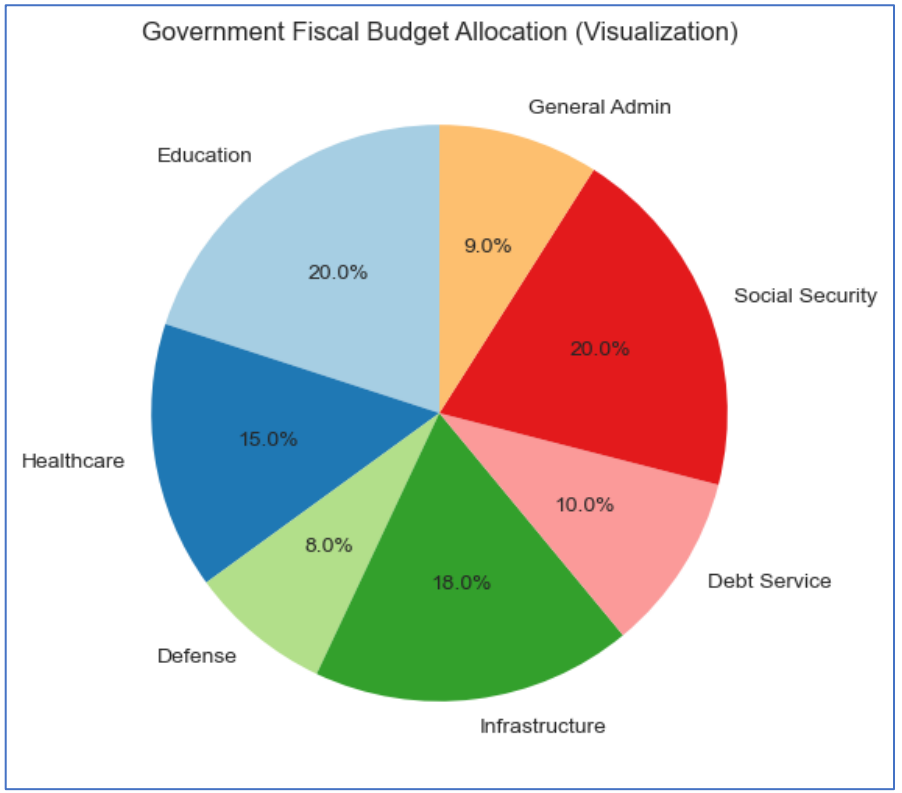
- **Fiscal Stimulus (*G*):** Used to fight recessions.
- **Multiplier Effect:** $1 of G > $1 of GDP.
- **Key Variable:** MPC (Marginal Propensity to Consume).
- **Leakages:** Imports and Savings reduce the effectiveness of stimulus.
- **Next:** Applying this to budget analysis.



Part 4: Data Visualization & Homework

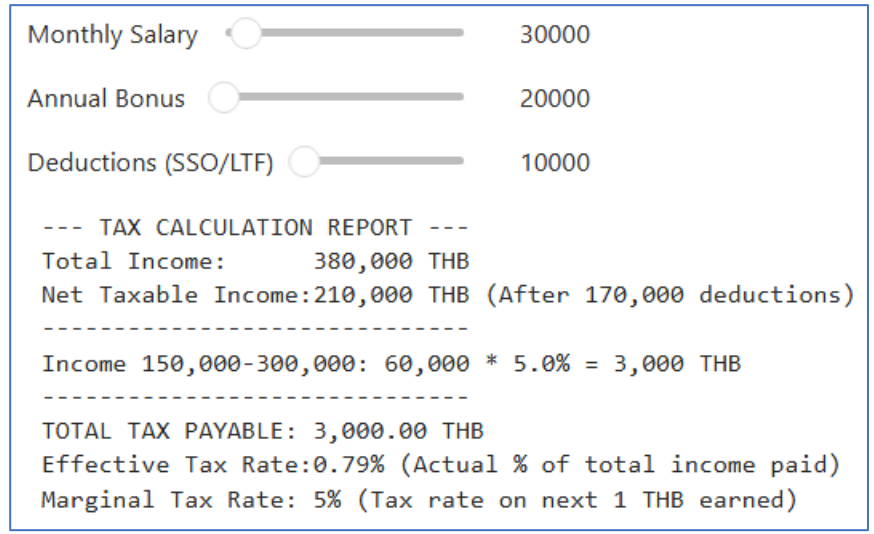
Visualizing the Government Budget

- Analyzing where taxpayer money goes.
- **Dataset:** Synthetic breakdown of a developing nation's budget.
- **Categories:** Education, Healthcare, Defense, Infrastructure, Debt Service.
- **Visualization:** Pie Chart.
- **Interpretation:** A budget reveals a nation's priorities. (e.g., Is Defense > Education?).



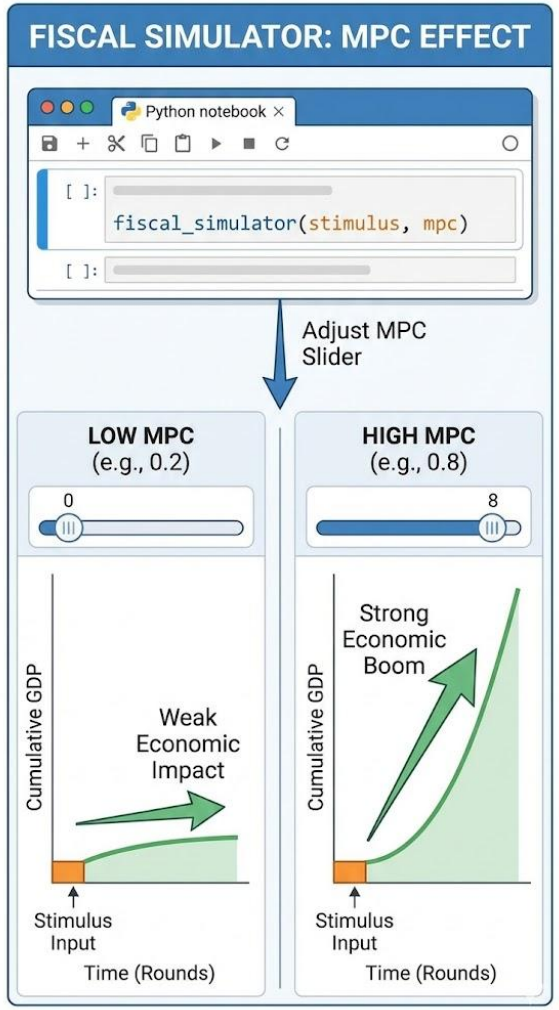
Notebook Walkthrough: Tax Calculator

- **Function:** calculate_thai_tax(salary, bonus, deductions).
- **Features:**
  - Handles standard deductions automatically.
  - Allows custom deductions (LTF/SSF).
  - Prints a detailed breakdown of tax paid per bracket.
- **Use Case:** Personal financial planning.



Notebook Walkthrough: Fiscal Simulator

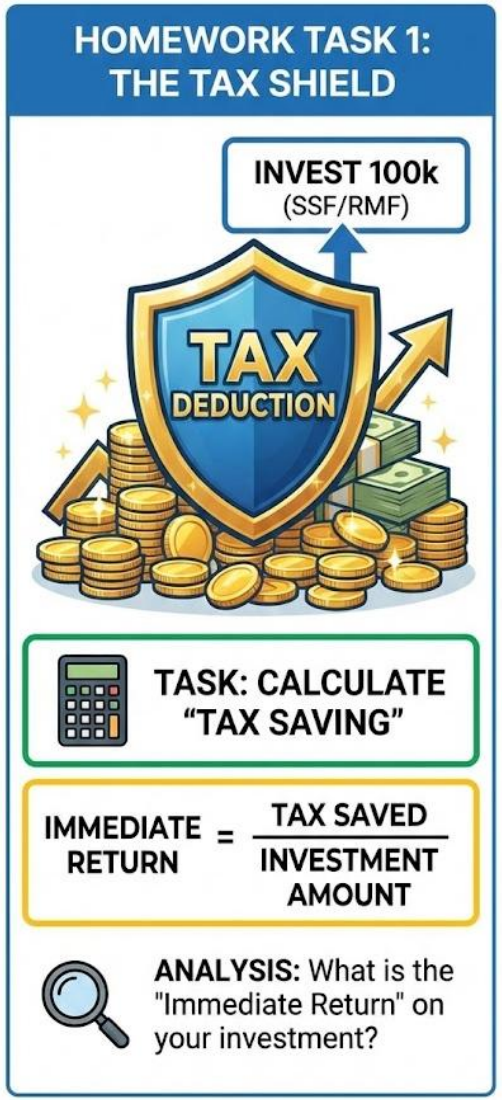
- **Function:** fiscal_simulator(stimulus, mpc).
- **Action:** Adjust the MPC slider.
- **Observation:**
  - Low MPC → The green line (Cumulative GDP) stays flat.
  - High MPC → The green line shoots up steeply.



Homework Task 1: The Tax Shield

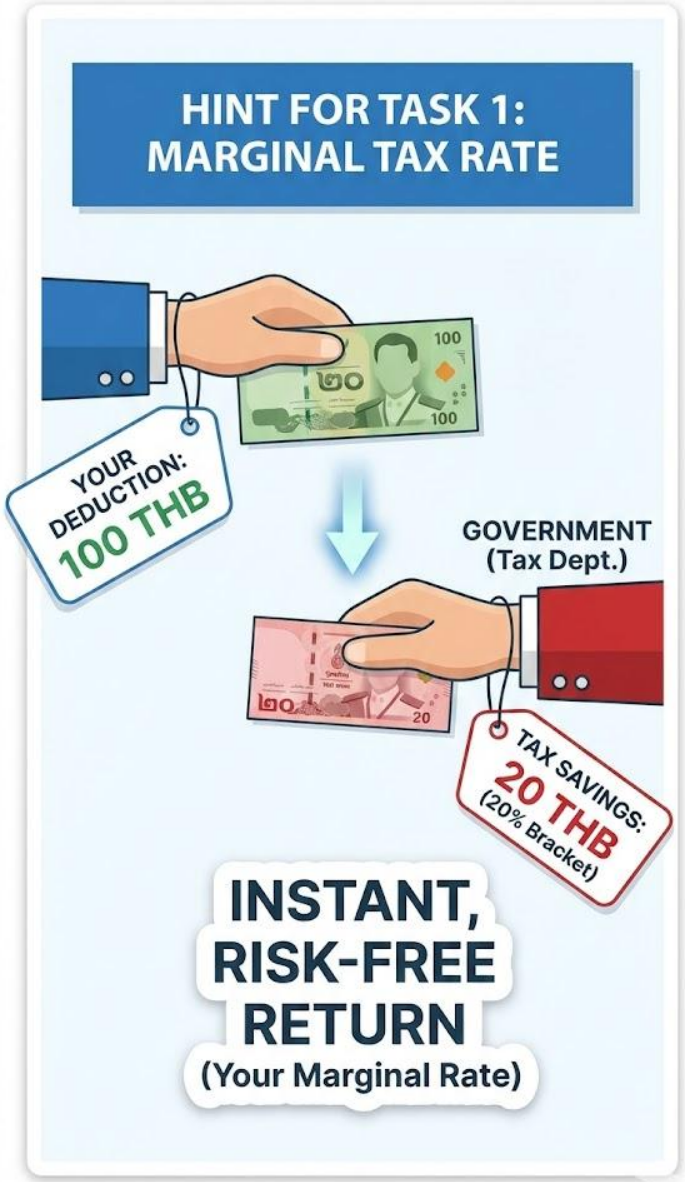
- **Context:** You earn 60k/month + Bonus.
- **Scenario A:** No investment. Calculate Tax.
- **Scenario B:** Invest 100k in a Tax-Deductible Fund (SSF/RMF).
- **Task:** Calculate the "Tax Saving."
- **Analysis:** Calculate the "Immediate Return."

$$Return = \frac{\text{Tax Saved}}{\text{Investment Amount}}$$



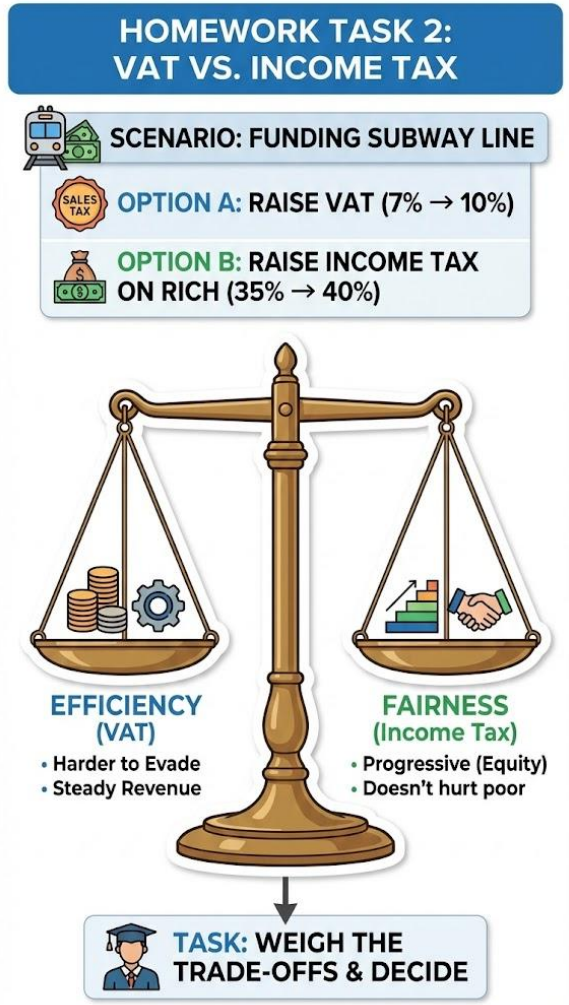
Hint for Task 1

- Identify your **Marginal Tax Rate** (The highest bracket you touch).
- If your top bracket is 20%, every 100 Baht you deduct saves you 20 Baht in tax.
- This is an instant, risk-free return on investment.



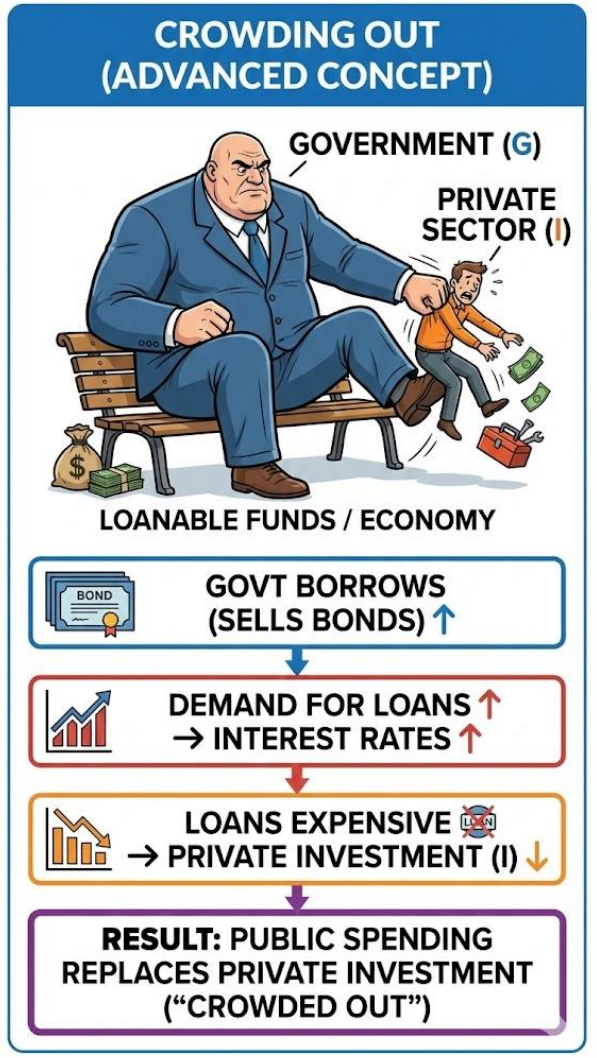
Homework Task 2: VAT vs. Income Tax

- **Scenario:** Funding a new subway line requires revenue.
- **Option A:** Raise VAT (7% → 10%).
- **Option B:** Raise Income Tax on the Rich (35% → 40%).
- **Trade-off:**
  - **Efficiency:** VAT is harder to evade and provides steady revenue.
  - **Equity:** Income Tax is progressive; VAT hurts the poor (Regressive).



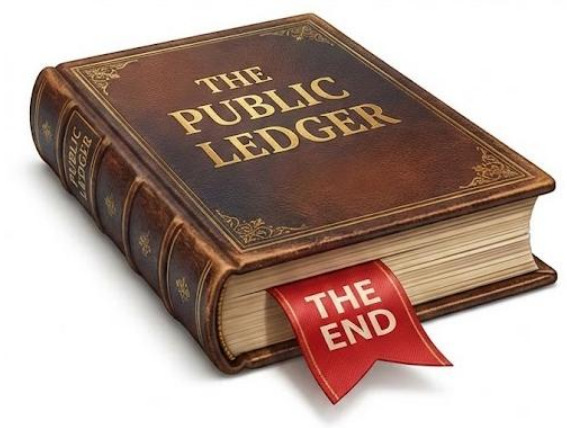
Crowding Out (Advanced Concept)

- **Warning:** If ***G*** increases too much, the government must borrow (sell bonds).
- **Effect:** This increases demand for loanable funds → Interest Rates rise.
- **Result:** Private Investment (***I***) falls because loans are expensive.
- **"Crowding Out":** Public spending replaces private investment.



Conclusion

- **Fiscal Policy:** The government's tool to steer the economy.
- **Taxation:** Must balance Revenue, Equity, and Efficiency.
- **Multiplier:** Spending creates ripples; consumption drives growth.
- **Next Session:** We wrap up the course.



🛞 **Fiscal Policy:** Government's tool to steer the economy.

⚖️ **Taxation:** Balance Revenue, Equity, and Efficiency.

💧 **Multiplier:** Spending creates ripples; consumption drives growth.

Let’s check out our Notebook

